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Multi-Criteria Clinic Allocation in Massachusetts to Reduce Access to Healthcare Disparities

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Abstract

Rural populations in the United States experience challenges in accessing healthcare facilities in ambulatory setting. While ambulatory healthcare facility location decisions have been widely studied, existing studies rarely use rurality and access to transportation together to inform equitable allocation of new facilities. We developed a novel optimization model to determine how to allocate a set of new clinics in a region to maximize coverage at the census tract level while considering rurality, lack of access to a vehicle, and existing clinics. Coverage is measured by the weighted size of vulnerable populations at the census tract level located within a predetermined distance from a new clinic. Weighting allows prioritization of census tracts with a: i) higher number of households with no access to a vehicle, and ii) higher level of rurality. Rurality is represented by the Rural-Urban Commuting Area codes. We applied the optimization model to allocate new clinics in Western Massachusetts, which includes 198 census tracts and 824,464 residents as of 2020. We used data from multiple publicly available datasets, and conducted sensitivity analyses on the distance threshold to new clinics and number of new clinics. Key findings indicate that Western Massachusetts has high rurality and a low number of clinics per census tract, thus highlighting the need for more clinics in the area. The proposed model serves as a useful tool for decision makers to select equitable clinic locations by operationalizing rurality and proportion of population with no access to a vehicle.

Keywords
Access to Transportation, Facility Location, Disparities, Linear Programming, Rurality

1. Introduction
Rural populations in the United States experience disparities related to availability of and access to healthcare facilities [1]. Since 2010, 138 rural hospitals have closed, and another 453 are identified as vulnerable to closure [2]. In addition, the COVID-19 pandemic exacerbated health disparities in rural communities by worsening existing issues (e.g., healthcare professional shortages) and by creating new challenges (e.g., suspension of outpatient services) [2]. There is a need for new clinics in ambulatory setting to provide care, particularly for rural populations who experience challenges related to access to care [3]. However, allocating new clinics in a given rural geographical region in an equitable way is associated with several challenges.

One of the challenges is that there is no consensus on how to define rurality. Various conventions are used such as Rural-Urban Commuting Area (RUCA) codes, Rural Urban Continuum Codes (RUCC), and U.S. census criteria [4]. Among these, the RUCA codes developed by the Economic Research Service (ERS) from U.S. Department of Agriculture (USDA) are most frequently used in literature [5]. These codes are classified from 1-10 based on size and direction of the primary commuting flows [6]. It remains to be determined how differences among rural communities change based on the definition of rurality and whether the discrepancy in definitions impact the measurement of disparities in health care access.

Another challenge is related to accurately quantifying the social vulnerability in a rural region. Social vulnerability, defined as the amount of injury, sickness or human loss that the community faces when perturbed by factors such as hazardous events or social disparities (e.g., race, gender, income, disability), should be considered in equitable clinic allocation decisions [7]. One way to quantify social vulnerability of a community is the Social Vulnerability Index (SVI), a metric developed by Centers for Disease Control and Prevention (CDC)/Agency for Toxic Substances and Disease Registry's (ATSDR) Geospatial Research, Analysis and Services Program (GRASP). SVI uses United States census data to determine the relative social vulnerability of every census tract. The SVI ranks 16 variables and groups them into four related themes (i.e., socioeconomic status, household composition, racial and ethnic minority status,

housing type and transportation) to rank every census tract and determine the most at-risk populations. One of the 16 components of SVI is percentage of households with no access to vehicle at the census tract level.

Since no access to a vehicle can worsen existing disparities, particularly in rural areas, considering both rurality and social vulnerability related to lack of transportation is crucial to incorporate equity into clinic location problems. Yet, connecting rurality and transportation related social vulnerability in facility allocation problems is an understudied area. Our research addresses this gap by modeling the interplay of rurality and access to transportation to make clinic allocation decisions. Specifically, we develop a novel optimization model to allocate p new clinics in 198 census tracts in Western Massachusetts to understand the influence of multiple factors such as rurality, number of existing facilities, and access to transportation in selecting the location of new clinics. In the remainder of this paper, we provide a quick review of the types of facility location models and existing literature on social vulnerability and facility location problems, discuss the data sources and the proposed optimization model, and conclude with results and future steps.

2. Literature Review

2.1 Types of Facility Location Models

Facility location problems are well studied in literature and often modeled as mixed integer programs [8]. Commonly used models are categorized as: (1) median (2) center, and (3) coverage models. Median models minimize the weighted sum of the distances between p facilities to be opened and a set of demand points. Center models minimize the maximum distance between any demand and its nearest facility. Coverage models seek to ensure that every customer is covered by a certain facility such that the distance between them and the facility is lower than a chosen threshold. In this study, a maximum coverage model was chosen to maximize the size of vulnerable populations located within a predetermined distance from a new clinic using a given number of clinics to be allocated in a specified rural region.

2.2 Social Vulnerability and Facility Location Problems

Different aspects of social vulnerability have been utilized in facility location literature. For instance, Zhang et al. [9] integrated transportation difficulties in the context of humanitarian relief operations by considering both traditional transportation costs and deprivation costs, which estimate the welfare loss to victims due to delays in receiving emergency supplies through their scenario-based stochastic program. Chen et al. [10] studied the problem of unbalanced allocation of primary healthcare facilities in Shenzen, China. Their work highlights disparities based on socioeconomic factors and geographic distribution. Key findings indicate 567 potential locations for new facilities that increase the population covered within a 15-minute travel time by 63.46%. To address this, the authors propose a location-allocation optimization model for primary care centers. This model incorporates spatial accessibility, considering travel time and public transportation options, while also accounting for population density and socioeconomic vulnerability. While existing studies explored spatial features, care facility density, and social vulnerability in a given region, explicitly connecting rurality and lack of transportation, as two concepts exacerbating each other's impact on disparities in access to care [11], is an area yet to be explored. In this paper, we operationalize social vulnerability as two criteria embedded into the objective function, one being rurality measured by RUCA codes and the other being percentage of households with no vehicles as measured by one of the components of SVI.

3. Data Sources

There are 3 main data sources: (1) Healthcare facility data from Massachusetts Department of Public Health (MA DPH), (2) SVI data from Centers for Disease Control and Prevention Agency for Toxic Substances and Disease Registry (CDC/ATSDR), and (3) RUCA codes from US Department of Agriculture Economic Research Service (USDA ERS).

- The healthcare facility data is the list [13] of all healthcare facilities licensed or certified by the Division of Health Care Facility Licensure and Certification from MA DPH (released on June 7, 2023). Only clinics in ambulatory setting are considered in this study.
- The SVI 2020 [14] is a dataset at the census tract level which includes 159 variables including important features such as Federal Information Processing Standards (FIPS) codes, state and county names, population estimates and 16 variables that correspond to social vulnerability categorized into 4 general themes: (1) socioeconomic status (2) household characteristics (3) racial and ethnic minority status, and (4) housing type and transportation.
- Finally, the RUCA dataset [15] is provided by the USDA ERS based on decennial updates and the 2006-2010 American Community Survey (ACS) at the census tract level. The data is nationwide and includes FIPS codes,

state and county information, primary and secondary RUCA codes and tract characteristics.

4. Methodology

The maximum covering location problem (MCLP) is a mathematical optimization problem concerned with selecting locations for facilities to maximize the coverage of populations or locations over a specified region. In this study, we developed a MCLP model for choosing locations (i.e., census tracts) for new clinics in Western Massachusetts.

4.1 Model Parameters and Decision Variables

The model parameters are listed below:

- r_i: rurality of census tract [1-10] $i \in I$ measured by RUCA codes where 1.0 is the metropolitan area core indicating primary flow within an urbanized area (UA) and 10.0 is the rural areas: primary flow to a tract outside a UA or urban cluster. The parameter values represent primary RUCA code values.
- e_i: Proportion of households (between 0 and 100%) with no access to a vehicle in census tract i obtained from a column in the SVI dataset
- w_i: Total population in census tract i obtained from a column in the SVI dataset
- N_j: Binary number [0 or 1] indicating whether there are any existing clinics in census tract j derived from the HCF dataset. N_j is 1 if a census tract has at least one existing clinic, and 0 otherwise.
- d_{ij}: Distance of centroid of census tract i from the centroid in census tract j, measured in miles
- A: Maximum distance within which a population in census tract i can be assigned to clinic in census tract j, measured in miles

The decision variables are listed below:

- x_{ij}: Binary variable indicating whether census tract i is covered by a clinic in census tract j
- y_j: Binary variable indicating whether a clinic is placed in census tract j
- z_i: Binary variable indicating whether census tract i is covered by a clinic

4.2 Optimization Model

Suppose that there exist $i \in I$ census tracts in Western Massachusetts, and $j \in J$ possible facility locations, then the amount of vulnerable households is given by $w_i e_i$. The goal of this model is to maximize the total number of covered vulnerable households in 198 census tracts of Western Massachusetts including 4 counties (Hampden, Berkshire, Hampshire, Franklin). This is achieved by selecting the locations (census tracts) for p new clinics. The objective function is given by Z as described by 1.

$$Z = \left\{ \max \sum_{\forall i \in I} (r_i e_i w_i) z_i \right\} \tag{1}$$

$$\text{s.t.} \quad \sum_{j \in J} x_{ij} = z_i \quad \forall i \in I \tag{2}$$

$$\sum_{j \in J} x_{ij} \leq y_j \tag{3}$$

$$\sum_{j \in J} y_j \leq p \tag{4}$$

$$y_j + N_j \leq 1 \quad \forall j \in J \tag{5}$$

$$d_{ij} x_{ij} \leq A y_j \quad \forall i, j \tag{6}$$

$$x_{ij}, y_i, z_i \in \{0, 1\} \tag{7}$$

Constraint 2 ensures that if a census tract i is covered ($z_i = 1$) then the census tract i must be assigned to exactly one new clinic in j. Constraint 3 ensures that an assignment from census tract i to clinic j can only happen if there is a clinic in location j. Constraint 4 ensures that the number of new clinics allocated are limited to p. Constraint 5 ensures that a clinic is added only to those census tracts that do not have existing clinics. Constraint 6 ensures that a

census tract is considered covered by a clinic in location j if census tract i is within the threshold distance A of clinic j. Constraint 7 indicates that all the decision variables are binary.

The distance calculation was performed using the Haversine formula to calculate the geodesic distance between two census tracts by using the latitude and longitude measurements, referred as lat and lon in Equation (8), of the centroids of each of the census tracts as shown in Equation (8). The minimum distance between the census tracts considered in this study is 0.38 miles, median distance is 20.1 miles, and the maximum distance between census tracts is 60.2 miles.

$$d = 2 \cdot R \cdot \sin\left(\frac{\Delta \text{lat}}{2}\right) + \cos(\text{lat}_1) \cdot \cos(\text{lat}_2) \cdot \sin\left(\frac{\Delta \text{lon}}{2}\right) \tag{8}$$

4.3 Model Assumptions

This optimization model is built based on a set of assumptions as listed below:

- **Fixed number of facilities**: The model assumes a known number of clinics that need to be located.
- **Continuous coverage**: The model assumes that each clinic can cover a continuous region or area, providing service to all the people in that census tract.
- **Uniform coverage quality**: The model assumes that the coverage provided by each clinic is uniform within its coverage area. In reality, coverage quality may vary depending on factors like distance, travel time, or other service-specific attributes.
- **Orthodromic distance metric**: The Orthodromic distance is used as the metric to calculate the distance between clinics and census tracts. This may not reflect the actual travel or transport costs accurately.
- **No facility capacity constraints**: The model does not consider clinic capacity constraints.
- **Single clinic coverage**: Each geographic census tract is exclusively served by one and only one clinic, without overlap or shared service areas between different clinics.

5. Results

The study included a total of 147 existing clinics located in 4 counties corresponding to 198 census tracts in Western Massachusetts. Table 1 shows the location of existing clinics at the county level and clinic to tract ratio (i.e., number of existing clinics divided by total census tracts). Results show that the number of existing clinics is less than the number of census tracts in all counties. Additionally, the table shows the average number of households with no access to vehicle per county and the average rurality of the counties on a scale of 1-10. Average number of households with no access to vehicle per county refers to the average value of the number of households in every census tract of the county that do not have access to a vehicle. Similarly, average rurality denotes the average RUCA codes of the census tracts in each county. Results indicate that Berkshire and Franklin counties are more rural than Hampden and Hampshire counties. Furthermore, Hampden county has the most number of households without a vehicle whereas Berkshire county has the least number of households without a vehicle.

Table 1: Descriptive Results of Model Census Tracts in Western Massachusetts

County	**Total Census Tracts**	**Total Existing Clinics**	**Clinic to Tract Ratio**	**Average number of households without a vehicle**	**Average RUCA Codes**
Berkshire	40	26	0.65	124.50	3.53
Franklin	19	11	0.57	126.42	3.91
Hampden	104	85	0.82	227.41	1.04
Hampshire	35	25	0.71	129.97	1.22

Based on the MCLP model, the results in Table 2 show the optimal location of clinics that can be placed in the 198 census tracts of Western Massachusetts given social vulnerability characteristics as well as the percentage of the

Table 2: MCLP model results

Acceptable distance (miles)	p	Facilities Chosen	RUCA codes covered
5	1	1: Franklin	4
5	2	2: Franklin	4
5	3	2: Franklin, 1: Berkshire	4 and 10
5	4	2: Franklin, 2: Berkshire	4 and 10
5	5	3: Berkshire, 2: Franklin	4 and 10
5	10	3: Berkshire, 3: Hampden, 2: Franklin, 2: Hampshire	1, 1.13, 4 and 10
10	1	1: Franklin	4
10	2	2: Franklin	4
10	3	2: Franklin, 1: Berkshire	4 and 10
10	4	2: Franklin, 2: Berkshire	4 and 10
10	5	3: Berkshire, 2: Franklin	4 and 10
10	10	3: Berkshire, 3: Hampden, 2: Franklin, 2: Hampshire	1, 1.13, 4 and 10

population covered by these added facilities. We varied A, the threshold distance, and p, the number of new clinics, to allocate clinics in census tracts that do not have any existing facilities. The results showed that the A value did not impact the final allocation of clinics for the given problem. However, as p was increased, the new clinic allocation changed. Initially, all new clinics were allocated to census tracts in Franklin county, and as p increased, some clinics were allocated to Berkshire county. It is interesting to note that the census tracts in Franklin county are on average more rural than Berkshire county as observed in Table 1. As p exceeded 5, the new clinics were placed in additional counties such as Hampden and Hampshire with the majority being placed in Hampden county which had more households on average without access to a vehicle as observed in Table 1.

6. Conclusion

This study provides evidence that there is an unequal distribution of primary healthcare clinics in Western Massachusetts as shown by the clinic to census tract ratio and highlights how rurality and transportation related social vulnerabilities play a role in enhancing access to clinics disparities. The results showed the impact of these vulnerability factors and the number of clinics on facility allocation in the census tracts of Western Massachusetts. The study has several limitations. Firstly, the study was performed on a subset of counties of Massachusetts focusing on Western Massachusetts and can be extended to cover the entire state. Secondly, the factors that influence healthcare access inequities can be further expanded to not only rurality and transportation related vulnerability but to the income of the population in each census tract, social support in accessing care etc. Furthermore, the distance calculation does not follow the road network. It is especially problematic in Western Massachusetts where the wetlands and protected areas make it such that straight line travel is not possible. Additionally, the model does not take into account the capacity of these newly placed clinics as well as the time it takes to reach them. In the real world, these factors can further limit the placement of new clinics to a subset of available census tracts. Further research can also allow the optimization model to be more precise about the placement of the types of primary care facilities (clinics, clinic satellites, mobile/portable clinics, mobile/portable clinic satellites) in each of these census tracts.

In conclusion, our findings can assist decision makers in assigning clinics based on rurality, proportion of households with no access to a vehicle and number of existing clinics to improve the access to clinics, particularly for rural communities. Future work will consider the impact of other social vulnerability factors as well as travel time, clinic capacity to target vulnerable areas throughout the state and the rest of the country.

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